

संभव BATCH - CRASH COURSE

Geography: Atmospheric Structure, Composition, Pressure & Temperature Belts

Expert Guidance: Vipulendra Tripathi Sir

SSC CGL Level Exams | 50 Selected Practice Questions (30 Moderate + 20 Advanced) | Pure Practice Set
(No Key Included)

PART - A: CONCEPTUAL & MODERATE LEVEL QUESTIONS

Q1. Which layer of the atmosphere contains the highest concentration of ozone gas?

- (A) Mesosphere
- (B) Stratosphere
- (C) Thermosphere
- (D) Troposphere

SSC CGL 2021

Q2. The atmospheric layer closest to the Earth's surface where almost all weather phenomena occur is the:

- (A) Exosphere
- (B) Stratosphere
- (C) Ionosphere
- (D) Troposphere

SSC CHSL 2022

Q3. What is the standard value of the Normal Lapse Rate of temperature in the troposphere?

- (A) 7.5°C per 1,000 meters
- (B) 6.5°C per 1,000 meters
- (C) 5.5°C per 1,000 meters
- (D) 4.5°C per 1,000 meters

SSC CPO 2023

Q4. Which atmospheric layer reflects radio waves back to the Earth's surface, enabling long-distance communication?

- (A) Mesosphere
- (B) Stratosphere
- (C) Exosphere
- (D) Ionosphere

SSC MTS 2022

Q5. The line joining places of equal atmospheric pressure at sea level is called an:

- (A) Isohyet
- (B) Isobar
- (C) Isohel
- (D) Isotherm

SSC CGL 2020

Q6. The Intertropical Convergence Zone (ITCZ) is characterized by a permanent belt of:

- (A) High pressure and strong westerlies
- (B) Low pressure and high-velocity katabatic winds
- (C) High pressure and descending air masses
- (D) Low pressure and calm winds

SSC Selection Post Phase X

Q7. Which permanent pressure belt is also widely known as the 'Horse Latitudes'?

- (A) Equatorial Low Pressure Belt
- (B) Polar High Pressure Belt
- (C) Subpolar Low Pressure Belt
- (D) Subtropical High Pressure Belt

SSC CHSL 2021

Q8. The deflection of planetary winds due to the Earth's self-rotation is a result of:

- (A) Coriolis Force
- (B) Pressure Gradient Force
- (C) Gravitational Pull
- (D) Centrifugal Force

SSC CPO 2020

Q9. Winds that blow continuously from the Subtropical High Pressure Belt towards the Equatorial Low Pressure Belt are termed:

- (A) Westerlies
- (B) Trade Winds
- (C) Polar Easterlies
- (D) Monsoon Winds

SSC CGL 2019

Q10. The boundary layer that physically separates the troposphere from the stratosphere is officially termed the:

- (A) Mesopause
- (B) Magnetopause
- (C) Tropopause
- (D) Stratopause

SSC CHSL 2022

Q11. Which of the following gases is the most abundant by volume in the dry composition of Earth's atmosphere?

- (A) Carbon Dioxide
- (B) Nitrogen
- (C) Argon
- (D) Oxygen

SSC Selection Post Phase XI

Q12. What is the absolute order of abundance of noble gases in the dry atmosphere?

- (A) Argon > Helium > Neon > Xenon
- (B) Helium > Neon > Argon > Xenon
- (C) Argon > Neon > Helium > Krypton
- (D) Neon > Argon > Helium > Krypton

SSC MTS 2021

Q13. In which atmospheric layer do meteors completely burn up upon entering from outer space?

- (A) Troposphere
- (B) Mesosphere
- (C) Thermosphere
- (D) Stratosphere

SSC CGL Tier-I 2022

Q14. The phenomenon of 'Anabatic Winds' refers to trade movements where air moves:

- (A) Up a mountain slope during the day
- (B) Down a valley slope during the night
- (C) Cyclonically around a low-pressure core
- (D) Horizontally across subpolar margins

SSC CPO 2021

Q15. Which instrument is universally utilized by meteorologists to record precise changes in atmospheric pressure?

- (A) Anemometer
- (B) Hygrometer
- (C) Thermometer
- (D) Barometer

SSC MTS 2022

Q16. The subpolar low-pressure belts are primarily dynamically formed due to:

- (A) Thermal contract indices over maritime channels
- (B) Sustained accumulation of cold descending polar air masses
- (C) The Earth's rotation and centrifugal forces displacing air
- (D) Intense solar insolation heating the subpolar margins

SSC CHSL 2020

Q17. The unit commonly used to express atmospheric pressure in professional meteorological maps is:

- (A) Pascal / Newton
- (B) Millibar / Hectopascal
- (C) Millimeter of Mercury exclusively
- (D) Torr / Square Inch

SSC CGL 2021

Q18. Westerlies in the Southern Hemisphere blow with exceptional velocity and are termed 'Roaring Forties' at 40°S because:

- (A) The Coriolis force is exponentially stronger in the Southern Hemisphere
- (B) The polar high-pressure gradient increases during summers
- (C) The subpolar lows undergo severe seasonal contraction
- (D) There is an absolute dominance of vast open oceans with minimal land interference

SSC CPO 2022

Q19. What is the primary heat transfer process responsible for warming the lower layers of the troposphere directly from the Earth's surface?

- (A) Sub-surface geothermal induction
- (B) Advection exclusively
- (C) Latent condensation loop
- (D) Conduction and Radiation

SSC MTS 2021

Q20. The beautiful atmospheric display of colored lights known as Aurora Borealis occurs within the:

- (A) Mesosphere
- (B) Exosphere
- (C) Stratosphere
- (D) Ionosphere / Thermosphere

SSC Selection Post Phase XI

Q21. The percentage composition of Carbon Dioxide (CO_2) by volume in dry air is approximately:

- (A) 0.003%
- (B) 0.04%
- (C) 0.12%
- (D) 0.93%

SSC CGL 2022

Q22. Which gas acts as a primary variable greenhouse component, decreasing rapidly with increasing altitude?

- (A) Water Vapor
- (B) Nitrogen
- (C) Argon
- (D) Oxygen

SSC CHSL 2022

Q23. The D, E, and F layers are sub-structural ionizational horizons belonging exclusively to the:

- (A) Stratosphere
- (B) Troposphere
- (C) Ionosphere
- (D) Mesosphere

SSC CPO 2021

Q24. A state of 'Temperature Inversion' implies that as altitude increases, the ambient temperature:

- (A) Remains perfectly constant
- (B) Drops instantly to absolute zero
- (C) Decreases at the normal lapse rate
- (D) Increases

SSC Selection Post Phase IX

Q25. Which wind belt dominates the mid-latitude zone between 35° and 60° latitudes in both hemispheres?

- (A) Northeast Trade Winds
- (B) Westerlies
- (C) Polar Easterlies
- (D) Doldrums

SSC CGL Tier-I 2021

Q26. The region of calm winds and variable light breezes centered directly over the equatorial low-pressure trough is known as the:

- (A) Doldrums
- (B) Subpolar Front
- (C) Roaring Forties
- (D) Horse Latitudes

SSC CHSL 2020

Q27. What is the approximate vertical height up to which the troposphere extends over the Equatorial regions?

- (A) 16 to 18 km
- (B) 6 to 8 km
- (C) 25 to 30 km
- (D) 50 km

SSC MTS 2022

Q28. Commercial jet aircraft prefer to cruise in the lower stratosphere primarily to evade:

- (A) Turbulent convective weather and clouds of the troposphere
- (B) The intense high-pressure resistance of polar cells
- (C) The strong frictional drag of the ionosphere
- (D) The extreme freezing effects of the mesopause

SSC CGL 2022

Q29. The Stratosphere is structurally characterized by a vertical temperature profile that:

- (A) Fluctuates erratically every hundred meters
- (B) Increases with height due to UV absorption by ozone
- (C) Decreases with height due to greenhouse cooling
- (D) Remains zero throughout its entire boundary length

SSC CPO 2023

Q30. The absolute lowest temperature naturally recorded within the Earth's atmospheric profile is located at the:

- (A) Tropopause
- (B) Thermopause
- (C) Stratopause
- (D) Mesopause

SSC Selection Post Phase XI

PART - B: ADVANCED MULTI-STATEMENT & ANALYTICAL QUESTIONS

Q31. Consider the following statements regarding the structural composition of the atmosphere:

1. Water vapor is a highly variable gas that decreases rapidly with altitude and drops from 4% in warm humid tropics to less than 1% in dry polar regions.
2. Dust particles and sea salt acts as hygroscopic condensation nuclei around which water vapor condenses to form clouds.
3. Carbon dioxide is transparent to incoming solar radiation but completely opaque to outgoing terrestrial longwave radiation.

Which of the statements given above are correct?

- (A) 1 and 3 only
- (B) 1 and 2 only
- (C) 2 and 3 only
- (D) 1, 2 and 3

SSC CGL Tier-II 2022

Q32. Assertion (A): The Troposphere has a significantly greater vertical height at the Equator (approx 18 km) compared to the Poles (approx 8 km).

Reason (R): Strong convective currents driven by intense solar heating transport heat and air masses to much higher altitudes over equatorial bands.

Select the correct code:

- (A) A is true but R is false.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) Both A and R are true and R is the correct explanation of A.
- (D) A is false but R is true.

SSC CGL Tier-II 2023

Q33. Match List-I (Atmospheric Horizon Layer) with List-II (Dominant Physical Property / Transition Invariant):

List-I

- A. Troposphere
- B. Stratosphere
- C. Mesosphere
- D. Ionosphere

List-II

- 1. Vertical temperature increase driven by UV filtration loops
- 2. Host to high-frequency reflection horizons like Kennelly-Heaviside layers
- 3. Dominance of strong convective mixing and standard Normal Lapse Rate
- 4. Reaches the absolute minimum temperature threshold near its upper boundary

Select the correct matching matrix sequence:

- (A) A-3, B-1, C-4, D-2
- (B) A-1, B-3, C-2, D-4
- (C) A-3, B-4, C-1, D-2
- (D) A-2, B-1, C-4, D-3

SSC CPO Tier-II 2021

Q34. Consider the following statements regarding the global Planetary Wind and Pressure Systems:

- 1. The Subtropical High-Pressure Belts are dynamically induced systems caused by the subsidence of air columns cooling from higher altitudes.
- 2. The Coriolis force operates at a maximum threshold at the Equator and decreases gradually to zero at both geographical poles.
- 3. Planetary trade winds are deflected to the right of their path in the Northern Hemisphere and to the left in the Southern Hemisphere due to Ferrel's Law.

Which of the statements given above are correct?

- (A) 1, 2 and 3
- (B) 1 and 2 only
- (C) 1 and 3 only
- (D) 2 and 3 only

SSC CGL Tier-II 2023

Q35. Which of the following statements is NOT correct regarding the Ionosphere and long-distance communication?

- (A) The concentration of free electrons decreases systematically as one climbs into higher horizons of the ionosphere.
- (B) The F-layer (Appleton layer) is the highest reflecting layer and remains operational during both day and night.
- (C) The D-layer is highly effective at reflecting low-frequency radio waves but completely disappears during night hours.
- (D) Solar flares trigger sudden ionospheric disturbances that can completely disrupt high-frequency radio transits.

SSC CHSL Tier-II 2022

Q36. Consider the following statements regarding the structural properties of atmospheric Pressure Belts:

1. The Equatorial Low-Pressure Belt is purely a thermally induced belt generated by intense year-round solar insulation.
2. Subpolar Low-Pressure Belts are zones of convergence between warm westerlies and cold polar easterlies, forming frontogenesis centers.
3. High-pressure cells over the polar caps are purely dynamic consequences of the Earth's high-speed diurnal rotation.

Which of the statements given above are correct?

- (A) 2 and 3 only
- (B) 1 and 2 only
- (C) 1, 2 and 3
- (D) 1 and 3 only

SSC Selection Post Phase XI

Q37. Assertion (A): The annual shifting of the pressure belts creates distinct seasonal reversal regimes of winds across monsoon regions.

Reason (R): Pressure belts shift northward during the summer solstice and southward during the winter solstice following the apparent movement of the sun.

Select the correct code:

- (A) A is true but R is false.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) A is false but R is true.
- (D) Both A and R are true but R is NOT the correct explanation of A.

SSC CGL Tier-II 2022

Q38. Match List-I (Specialized Planetary Wind Name) with List-II (Latitudinal Zone Constraints):

List-I

- A. Doldrums**
- B. Horse Latitudes**
- C. Roaring Forties**
- D. Furious Fifties**

List-II

- 1. 30° to 35° North and South latitudes (Subtropical Highs)**
- 2. 50°S latitude containing high-velocity maritime westerlies**
- 3. 5°N to 5°S latitude centered on the equatorial thermal trough**
- 4. 40°S latitude showcasing continuous ocean-driven gale vectors**

Select the correct matching sequence:

- (A)** A-4, B-1, C-2, D-3
- (B)** A-3, B-1, C-4, D-2
- (C)** A-1, B-3, C-4, D-2
- (D)** A-3, B-4, C-1, D-2

SSC CGL Tier-II 2023

Q39. Consider the following statements regarding the phenomenon of 'Temperature Inversion':

- 1. Surface inversion is highly favored by long winter nights, a clear unclouded sky, and perfectly still calm air.**
- 2. Inversion of temperature completely stabilizes the lower atmosphere, effectively suppressing any vertical convective cloud mixing.**
- 3. Valley inversion occurs when dense cold air drains downslope under gravity to settle at the valley floor, displacing warmer air upward.**

Which of the statements given above are correct?

- (A)** 1 and 2 only
- (B)** 1, 2 and 3
- (C)** 1 and 3 only
- (D)** 2 and 3 only

SSC CPO Tier-II 2022

Q40. Which of the following configurations represent the correct descending order of gases by volume in the Homosphere section of the atmosphere?

1. Nitrogen (N_2)
2. Oxygen (O_2)
3. Argon (Ar)
4. Carbon Dioxide (CO_2)

Select the correct structural sequence:

- (A) $2 > 1 > 3 > 4$
- (B) $1 > 2 > 4 > 3$
- (C) $1 > 3 > 2 > 4$
- (D) $1 > 2 > 3 > 4$

SSC Selection Post Phase XI

Q41. Consider the following statements regarding the 'Coriolis Force' mechanics:

1. It is an apparent force caused purely by the Earth's rotation, operating perpendicular to the wind direction axis.
2. It directly alters the absolute structural velocity or speed of the planetary wind vectors.
3. The value of Coriolis deflection is directly proportional to the sine of the latitude angle.

Which of the statements given above is/are correct?

- (A) 1 only
- (B) 1, 2 and 3
- (C) 1 and 3 only
- (D) 2 and 3 only

SSC CGL Tier-II 2022

Q42. Assertion (A): Cyclonic systems in the Northern Hemisphere display a counter-clockwise (anticlockwise) circulation pattern.

Reason (R): Winds blowing inward toward the central low-pressure core are deflected to their right by the Coriolis force.

Select the correct options:

- (A) A is false but R is true.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) A is true but R is false.
- (D) Both A and R are true but R is NOT the correct explanation of A.

SSC CGL Tier-II 2023

Q43. Match List-I (Atmospheric Pause Boundary) with List-II (Approximate Vertical Elevation Limit):

List-I

- A. Tropopause (Equatorial mean)**
- B. Stratopause**
- C. Mesopause**
- D. Heterosphere base boundary**

List-II

- 1. Approx 50 kilometers**
- 2. Approx 18 kilometers**
- 3. Approx 100 kilometers**
- 4. Approx 80 to 85 kilometers**

Select the correct sequence:

- (A)** A-2, B-4, C-1, D-3
- (B)** A-1, B-2, C-4, D-3
- (C)** A-2, B-1, C-4, D-3
- (D)** A-3, B-1, C-4, D-2

SSC CPO Tier-II 2021

Q44. Consider the following statements about the 'Mesosphere' layer:

- 1. Temperature drops continuously with increasing altitude within this layer, reaching around -100°C near its upper limit.**
- 2. Noctilucent clouds are rare, ice-crystal-studded formations that develop within this layer near polar margins during summer.**
- 3. It lacks any significant chemical compounds or permanent dry gases, being composed entirely of free electrons.**

Which of the statements given above are correct?

- (A)** 1 and 3 only
- (B)** 1, 2 and 3
- (C)** 2 and 3 only
- (D)** 1 and 2 only

SSC CHSL Tier-II 2022

Q45. Which of the following statements is correct regarding the 'Hadley Cell' circulation pattern?

- (A) Air rises at the Equatorial Lows, moves poleward in the upper troposphere, and descends at the Subtropical Highs.
- (B) Air descends vertically at the subpolar low-pressure margins and flows directly back towards the equatorial belt.
- (C) It operates as a completely closed cold thermal loop restricted within the polar stratospheric boundary.
- (D) It is a mid-latitude cell driven entirely by the dynamic mechanical interactions of the westerlies front.

SSC CGL Tier-II 2023

Q46. Consider the following statements about atmospheric composition and weight distribution:

1. Nearly 99% of the total mass of the atmosphere is confined within a vertical limit of 32 kilometers from the Earth's surface.
2. The chemical composition of the atmosphere regarding permanent dry gases remains perfectly uniform up to approximately 80 km.
3. Ozone is distributed uniformly across all layers of the atmosphere, with a constant concentration from sea level to the exosphere.

Which of the statements given above are correct?

- (A) 1, 2 and 3
- (B) 1 and 2 only
- (C) 2 and 3 only
- (D) 1 and 3 only

SSC CGL Tier-II 2022

Q47. Assertion (A): The subpolar low-pressure zones exhibit highly volatile, stormy cyclonic weather conditions, especially during winter months.

Reason (R): They mark the zone of frontogenesis where warm, moist subtropical maritime air masses collide with dense, freezing polar continental air masses.

Select the correct option:

- (A) Both A and R are true and R is the correct explanation of A.
- (B) A is true but R is false.
- (C) A is false but R is true.
- (D) Both A and R are true but R is NOT the correct explanation of A.

SSC CGL Tier-II 2023

Q48. Consider the following variables regarding the 'Ferrel Cell' circulation mechanics:

1. It operates in the mid-latitude zones between the Subtropical Highs and Subpolar Lows.
2. Unlike the Hadley and Polar cells, it is an indirect, mechanically driven cell rather than a purely thermally induced loop.
3. The surface expression of this cell constitutes the global westerly wind belts.

Which of the statements given above are correct?

- (A) 1 and 3 only
- (B) 1 and 2 only
- (C) 2 and 3 only
- (D) 1, 2 and 3

SSC CPO Tier-II 2022

Q49. Which of the following structural definitions accurately contrasts the 'Homosphere' and 'Heterosphere' layers?

1. The Homosphere extends from sea level up to roughly 80 km and has a highly uniform proportional blend of major permanent gases.
2. The Heterosphere lies above 80 km and is characterized by distinct, stratified gas layers organized by atomic weight (Molecular Nitrogen, Atomic Oxygen, Helium, Hydrogen).

Select the correct answer using codes below:

- (A) 1 only
- (B) Both 1 and 2
- (C) 2 only
- (D) Neither 1 nor 2

SSC CHSL Tier-II 2022

Q50. Consider the following statements about the 'Exosphere' layer:

1. It represents the uppermost horizon of the Earth's atmosphere, transitioning gradually into outer interplanetary space.
2. The air density is exceptionally low, and lightweight gases like Hydrogen and Helium dominate this zone.
3. Temperatures drop continuously within this layer down to absolute zero due to the lack of gas molecules to absorb heat.

Which of the statements given above are correct?

- (A) 1, 2 and 3
- (B) 1 and 2 only
- (C) 1 and 3 only
- (D) 2 and 3 only

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